From Household KKT to Growth

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ECON 803, Fall 2026

Questions only. This original teaching extension of Problem 1.6 closes household quantity and quality choice with competitive firms, land ownership and resource clearing, then adds population and human capital dynamics. It is a tractable classroom model inspired by Unified Growth Theory, not a reproduction of the complete Galor model.

These are the same questions as the textbook. Begin with the hand calculations and state a solution plan before using tools. Report the economic assumptions, the hand seed, one scaling step and an exact verification command. Numerical output alone is not a proof.

Optional background: Galor (2005), From Stagnation to Growth: Unified Growth Theory

Problem 16.A3. From household KKT to a land constrained growth economy

Suggested time: 90 minutes for the hand benchmarks. Total: 80 points. The optional program is not included in this time. This teaching case extends Problem 1.6 to competitive factor prices and a sequence of generations. It is inspired by Galor's quantity versus quality mechanism, not a reproduction of his full Unified Growth Theory.

There is one consumption good, whose price is one, fixed land $X > 0$, and a mass $N > 0$ of identical current adults with skill $h > 0$. A competitive constant returns firm rents land $Z \geq 0$ and raw adult labor hours $L \geq 0$ to produce $Y = AZ^{1-\alpha}(hL)^{\alpha}$, where $A > 0$ and $0 < \alpha < 1$. The land market must clear at $Z = X$. The wage $w > 0$ is paid per raw hour, and R is rent per unit of land. Every adult owns an equal land share and receives $\rho = RX/N$. Firm profit is distributed but is zero at a competitive supporting price.

Each adult has one unit of time. It chooses consumption $c > 0$, continuous fertility $n > 0$ and education time per child $e \geq 0$ to maximize $(1-\gamma)\log c + \gamma \log[nh_a(e)]$, where $0 < \gamma < 1$, $\tau > 0$, $a > 0$, $\bar{c} > 0$ and $h_a(e) = 1 + ae/(\tau + e)$. Total child time is $t = n(\tau + e)$. Constraints are $c + wt \leq w + \rho$, $t \leq 1$ and $c \geq \bar{c}$. Land rent is nonlabor income: child time sacrifices wages, not rent. Each household takes w and ρ as given. The logarithm excludes zero fertility in this model.

Part I. By hand

(a) Set $E = ne$ and $Q(n, E) = n + anE/(\tau n + E)$. State the transformed feasible set and write the complete KKT system, including the time bound, subsistence and the zero education entry inequality. Establish concavity and explain why a feasible KKT certificate gives a global optimum. [12]

(b) At fixed prices, write $I = w + \rho > \bar{c}$. Solve globally for education, child time and consumption, including all three possible bounds on child time and the equality case $a = 1$. Explain why replacing w by I in the cost of child time would be an economic error. [12]

(c) Define $B = Ah^{\alpha}(X/N)^{1-\alpha}$. A symmetric general equilibrium has $L = N(1 - t)$ and $Nc = Y$. Derive factor prices, zero profit and the resource identity. Find all positive equilibria for every $B > \bar{c}$, identify the subsistence threshold and verify the household KKT certificate at the equilibrium. Explain what happens to feasibility when $B \leq \bar{c}$. [16]

Dynamic benchmarks. Continue by hand. The optional program at the end checks these calculations; it does not replace the arguments. At the end of a generation all children become next generation adults, so $N' = nN$. Skill is inherited and augmented by education: $h' = hh_a(e)$. Land and A stay fixed, and land ownership is shared equally among the next cohort. Parents value child quantity and quality through the displayed utility but do not solve a forward looking dynamic program. Thus $n = 1$ means replacement of the current adult mass, not a literal demographic birth rate.

(d) Derive the one dimensional law for B'. Set $\alpha = 3/4$, $\gamma = 1/2$, $\tau = 1/4$ and $\bar{c} = 4$. With $a = 1/2$, find a subsistence steady state with replacement fertility and constant skill. Calculate the derivative of the local B map there and explain the stability conclusion, including its scope. [12]

(e) Start at that steady state and permanently raise a to 3. Calculate the impact education, child time, fertility and skill multiplier. Does the next generation cross the threshold into slack subsistence? Prove your comparison exactly. Compute the subsequent fertility and asymptotic growth factors of skill, adult mass, B and consumption. [18]

(f) Explain which feedbacks close the economy, why rents alter fertility incentives without being sacrificed per hour, and why the model does not deliver exact zero fertility or a full demographic transition. Distinguish an exogenous education return shock from endogenous technological progress, and distinguish the impact fertility response from its later level. [10]

Part II. Codex and verification

Optional tool extension. Keep $A = 8$, $X = h_0 = 1$ and all calibrated parameters fixed, with the permanent $a = 1/2$ to $a = 3$ change at date zero. Vary only initial adult mass $N_0 \in (0, 16)$, including the steady state and regime boundary populations obtained by hand and points immediately around them. State Seed, Fixed, Scaled and Risk before coding. Simulate twenty generations using independently implemented household and equilibrium routines. Check all budgets, time and subsistence bounds, raw labor and land clearing, zero profit, aggregate goods clearing, KKT signs and complementary products at every date. Compare the household formula with an independent bounded optimization at selected dates and state tolerances. Plot the paths and mark the subsistence switch. Report what is proved analytically, what is checked on this finite grid and what is only a maintained demographic assumption.